

ORIGINAL ARTICLE

Effects of potassium in reducing the radiocesium translocation to grain in rice

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Abstract

This study investigated the effects of potassium (K) on the behavior of cesium 137 (¹³⁷Cs) in hydroponically cultured rice plants (*Oryza sativa* L. “Nipponbare”), revealing that K supply has an influence on reducing ¹³⁷Cs translocation from leaves to ears as well as decreasing ¹³⁷Cs uptake by roots. When K was omitted from the culture solution and ¹³⁷Cs was supplied, the ¹³⁷Cs content in the ears was increased 6.7-fold compared with the K-sufficient plants, while the total ¹³⁷Cs content in the shoot part of the K-deficient rice was only 3-fold higher than that in the K-sufficient rice. The fraction of the ears in the total shoot ¹³⁷Cs content was indeed 2.5 times higher in the K-deficient rice (45%) than that in the K-sufficient rice (18%), demonstrating that K supplementation has an effect on ¹³⁷Cs translocation to the ears. Irrespective of the K condition, > 95% of the ¹³⁷Cs that accumulated in the ears was found to be absorbed from the culture solution before ear emergence. K supplementation after ear emergence had an influence on achieving the percentage of ¹³⁷Cs remobilization in the K-deficient rice close to that in the K-sufficient rice, although the effect on reducing the ¹³⁷Cs content in the ears could be limited. The results indicate the importance of maintaining an appropriate soil K concentration during the early growth stages to avoid ¹³⁷Cs contamination of rice grains.

Key word: potassium, radiocesium, rice, translocation, uptake.

INTRODUCTION

Soil contamination with radionuclides derived from nuclear weapons testing or accidental release from nuclear facilities has a long-term radiological impact on agriculture. After the accident at the Fukushima Daiichi nuclear power plant in 2011, radiocesium (¹³⁷Cs and ¹³⁴Cs) was the main artificially produced radionuclide in the east part of Japan, and ¹³⁷Cs requires special consideration because of its long half-life (30 years). To avoid the contamination of crops, which are the major source of radioactive contamination in the human body (Shaw *et al.* 1994), the mechanisms of ¹³⁷Cs transportation into rice (*Oryza sativa* L. “Nipponbare”) is a major concern in Japan.

¹³⁷Cs distribution in rice plants was studied in Fukushima during the autumn of 2011. Then, the rice plants grown in a paddy field where highly contaminated brown rice grains (approximately 500 Bq kg⁻¹) were uncommonly produced were demonstrated to contain higher ¹³⁷Cs levels in the newer leaves than in the older leaves, which was the opposite of what was found in rice plants grown in a paddy field which produced a grain with very low ¹³⁷Cs concentration Tanoi *et al.*, 2013; Nakanishi and Tanoi 2013, and also contrary to a previous report demonstrating that ¹³⁷Cs concentration was higher in old leaves than new leaves and panicles when grown in an experimental field replicating the typical paddy field (Tsukada *et al.* 2002). Based on these characteristic ¹³⁷Cs distribution in the rice plant, two hypotheses were conceived that may explain why brown rice containing higher ¹³⁷Cs concentrations was uncommonly produced in some paddy fields. First, there may have been seasonal variation in the environmental ¹³⁷Cs concentration, i.e., high ¹³⁷Cs levels may have been available in the paddy soil during the midsummer when upper leaves grow

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Received 16 March 2014.

Accepted for publication 20 July 2014.